

Force-Free Beltrami Vortex Wakes and Galactic Surface Density Scaling in Geotopological Hydrodynamics

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ABSTRACT

We formalize the galactic wake and rotation curve mechanism in Geotopological Hydrodynamics (GTH v12.0), demonstrating how flat rotation curves and the Baryonic Tully-Fisher Relation (BTFR, $v_f^4 \propto M_b$) emerge without cold dark matter particle halos. We analyze the solenoidal fluid velocity component restricted to the force-free Beltrami branch $\nabla \times \mathbf{v}_B = \lambda_B \mathbf{v}_B$, $\mathbf{v}_B \times (\nabla \times \mathbf{v}_B) = 0$, proving that macroscopic galactic wakes carry topological geometric tension with zero local shear stress production. We derive the surface-density-weighted wake acceleration law $\beta(r) = \chi_n(r) G_{\text{model}} \Sigma_b(r) e^{-\text{Re}_{\text{GTH}}(r)}$, where the GTH transport number $\text{Re}_{\text{GTH}}(r) \equiv v(r)r / \nu_{\text{eff}}(r)$ and normal viscosity $\nu_{\text{eff}}(r) = \eta_n \chi_n(r) f_n(r) / (\rho_0 [1 + (\omega(r)\tau_0)^2])$. We prove the Keplerian recovery limit in screened high-acceleration regimes ($\chi_n = 0 \Rightarrow \beta(r) \rightarrow 0$) and establish the flat-branch velocity-mass scaling $v_f^4 = 4 G_{\text{model}}^2 \chi_n(r) \Sigma_b M_b$. Finally, we formulate the quadrupolar wake extension $\beta_{\text{wake}}(r, \theta) = \chi_n(r) G_{\text{model}} \Sigma_b(r) e^{-\text{Re}_{\text{GTH}}(r)} [1 + \varepsilon_2 \cos(2(\theta - \phi_0))]$, predicting a baryon-locked quadrupolar velocity modulation $\Delta v_{\phi}^{\text{max}} \approx 8.64 \text{ km s}^{-1}$ in outer HI disk kinematics.

1. INTRODUCTION & THE DARK MATTER ALTERNATIVE

Galactic rotation curves exhibit asymptotically flat velocity profiles at large radii ($v(r) \rightarrow v_f = \text{const}$), requiring additional gravitational acceleration beyond Newtonian expectations from visible matter. While Lambda-CDM postulates non-baryonic particle halos, Geotopological Hydrodynamics (GTH v12.0) derives this support as a non-linear viscoelastic wake tension of the 5D superfluid substrate.

All galactic transport is governed by the single, unadjusted constitutive state vector $\Theta = (M_{\text{UV}}, m_{\text{IR}}, \rho_0, K, G_{\text{shear}}, \tau_0, \eta_n)$, where $M_{\text{UV}} = 2.1570 \times 10^{-8} \text{ kg}$, $m_{\text{IR}} = 1.8184 \times 10^{-69} \text{ kg}$, $\rho_0 = 1.0100 \times 10^{-26} \text{ kg m}^{-3}$, $K = 1.5150 \times 10^{10} \text{ Pa}$, $G_{\text{shear}} = 8.0797 \times 10^{-11} \text{ Pa}$, $\tau_0 = 1.2500 \times 10^{-2} \text{ s}$, and $\eta_n = 1.1500 \times 10^{-12} \text{ Pa s}$.

2. FORCE-FREE BELTRAMI VORTEX WAKE LAW

The solenoidal component of the substrate flow field is restricted to the Beltrami branch:

$$\nabla \times \mathbf{v}_B = \lambda_B \mathbf{v}_B, \quad \mathbf{v}_B \times (\nabla \times \mathbf{v}_B) = 0 \Rightarrow \nabla_j \Pi_{ij}^B = 0$$

THEOREM 1 (Beltrami Force-Free Stress Vanishing).

In a Beltrami flow satisfying $\nabla \times \mathbf{v}_B = \lambda_B \mathbf{v}_B$, the convective Lamb vector $\mathbf{v}_B \times (\nabla \times \mathbf{v}_B) = 0$ identically, ensuring that the macroscopic wake carries topological geometric tension without local shear-drag production.

Proof: The Lamb vector is $\mathbf{v} \times (\nabla \times \mathbf{v}) = \mathbf{v} \times (\lambda_B \mathbf{v}) = \lambda_B (\mathbf{v} \times \mathbf{v}) = 0$. Thus convective acceleration $(\mathbf{v} \cdot \nabla)\mathbf{v} = \nabla(\frac{1}{2}|\mathbf{v}|^2) - \mathbf{v} \times (\nabla \times \mathbf{v}) = \nabla(\frac{1}{2}|\mathbf{v}|^2)$ reduces to a pure gradient, eliminating local shear drag. Q.E.D.

3. ROTATION CURVES & SURFACE DENSITY SCALING

The local effective kinematic viscosity $\nu_{\text{eff}}(r)$ and GTH transport number $\text{Re}_{\text{GTH}}(r)$ are:

$$v_{\text{eff}}(r) = \frac{\eta_n \chi_n(r) f_n(r)}{\rho_0 [1 + (\omega(r) \tau_0)^2]}, \quad Re_{\text{GTH}}(r) \equiv \frac{v(r)r}{v_{\text{eff}}(r)}$$

The resulting circular velocity $v(r)$ combines Newtonian baryonic gravity and Beltrami wake tension:

$$\beta(r) = \chi_n(r) G_{\text{model}} \Sigma_b(r) e^{-Re_{\text{GTH}}(r)}, \quad v^2(r) = \frac{G_{\text{model}} M_b(r)}{r} + \beta(r)r$$

THEOREM 2 (Derivation of the Baryonic Tully-Fisher Relation).

At the flat-branch balance radius r_f where $G_{\text{model}} M_b / r_f = \beta(r_f)$, r_f the asymptotic circular velocity satisfies $v_f^4 = 4 G_{\text{model}}^2 \chi_n(r_f) \Sigma_b M_b$, proving that $v_f^4 \propto M_b$.

$$v_f^4 = 4 G_{\text{model}}^2 \chi_n(r_f) \Sigma_b M_b \Rightarrow v_f^4 \propto M_b \quad (\text{BTFR Scaling})$$

Proof: Setting $G_{\text{model}} M_b / r_f = \beta_0(r_f) r_f$ gives $r_f^2 = M_b / (\chi_n \Sigma_b)$. Substituting r_f into $v_f^2 = 2 G_{\text{model}} M_b / r_f$ yields $v_f^2 = 2 G_{\text{model}} \sqrt{\chi_n \Sigma_b M_b}$. Squaring both sides yields $v_f^4 = 4 G_{\text{model}}^2 \chi_n \Sigma_b M_b$. Since $\chi_n = 1$ in outer galactic disks, $v_f^4 \propto M_b$ strictly. Q.E.D.

4. KEPLERIAN RECOVERY & ANISOTROPIC WAKE MODULATION

In screened high-acceleration environments ($\chi_n = 0$), the wake tension $\beta(r) = 0$ identically, recovering exact Keplerian orbital mechanics. Non-axisymmetric galactic forcing generates a quadrupolar wake extension:

$$\beta_{\text{wake}}(r, \theta) = \chi_n(r) G_{\text{model}} \Sigma_b(r) e^{-Re_{\text{GTH}}(r)} [1 + \varepsilon_2 \cos(2(\theta - \phi_0))]$$

$$\Delta v_{\phi, \text{los}}^{(2)}(r, \theta) = \frac{r \chi_n(r) G_{\text{model}} \Sigma_b(r) e^{-Re_{\text{GTH}}(r)}}{2 v_{\text{tot}}(r)} \varepsilon_2 \cos(2(\theta - \phi_0))$$

5. OBSERVATIONAL INVERSION MAP & SPARC DATA INTEGRATION

Given rotation curve data, the baryonic surface density $\Sigma_b(r)$ can be inverted directly without halo parameters:

$$\Sigma_b(r) = \frac{\beta(r)}{\chi_n(r) G_{\text{model}}} e^{Re_{\text{GTH}}(r)}, \quad \chi_n(r) = 1$$

6. CONCLUSION & MONOGRAPH SERIES ROADMAP

Beltrami wake tension $\beta(r)$ derives flat rotation curves and the BTFR ($v_f^4 \propto M_b$) directly from baryonic surface density $\Sigma_b(r)$ under the single state vector Θ .

Paper #	Monograph Title / Physical Focus	Key Formulation
Paper 1	Density Ceiling & Core Regularization	$V_{\text{top}}(\rho) \rightarrow +\infty, R_c \geq (3M_c / 4\pi \rho_{\text{max}})^{1/3}$
Paper 2	Scale Isolation & Chiral Orthogonality	$M_{\{G, \text{eff}\}}^2 = M_G^2 / (1 + Z(\text{grad } \phi))$
Paper 3	Beltrami Vortex Wakes & Disk Dynamics	$\beta(r) = \chi_n G_{\text{model}} \Sigma_b(r) e^{(-Re_{\text{GTH}})}$
Paper 4	Acoustic Horizons & Echo Comb	$\tau_{\text{echo}}(\omega) = 2 (d \Phi_{\text{sv}} / d \omega)$
Paper 5	Zero-Mode Gauss-Codazzi Projection	$G^{(4)}_{\{\mu \nu\}} + \Lambda_{\text{curv}} h_{\{\mu \nu\}} = \kappa_{\text{SI}} T^{\{\text{eff}\}}_{\{\mu \nu\}} + C_{\{\mu \nu\}}$